BasisBridge

Multiple Coordinate System Converter - Experiment report

Aim: Transform vectors successively through different bases and verify the transformations.

Concepts: Basis and dimension - Change of basis - Matrix multiplication - Linear transformations - Inverse matrices

Vector space: 2D Input vector: [5 4]

Basis 1 (columns are basis vectors), det = 1

[1 0]
[0 1]
Coordinates in B1: [5 4]

Basis 2 (columns are basis vectors), det = -2

[1 1]
[1 -1]
Coordinates in B2: [4.5 0.5]

Basis 3 (columns are basis vectors), det = 2

[2 0]
[0 1]
Coordinates in B3: [2.5 4]

P01 = inverse(B1) x I
[1 0]
[0 1]
P21 = inverse(B2) x B1
[0.5 0.5]
[0.5 -0.5]
P32 = inverse(B3) x B2
[0.5 0.5]
[1 -1]
Reverse matrix inverse(P32)
[1 0.5]
[1 -0.5]
Reverse matrix inverse(P21)
[1 1]
[1 -1]
Reverse: [v]B2 = [4.5 0.5]

Reverse: [v]B1 = [5 4]

Reconstructed vector: [5 4]

Absolute coordinate errors: [0 0]

Maximum coordinate error: 0.000e+0

Verification: PASS - Transformation verified successfully.

Conclusion: Successive coordinate changes and reverse reconstruction were computed from the entered basis matrices.